

Extremal Combinatorics — Solutions

Problem 1. (Baltic Way 2018 Problem 10) Two players X and Y play a game as follows. There are n cards labelled $1, 2, \dots, n$ respectively. Player X first removes a card. Then player Y removes two cards with consecutive labels. After that player X removes three cards with consecutive labels. Finally, player Y removes four cards with consecutive labels. What is the smallest value of n for which player Y can guarantee to complete both moves no matter how player X plays?

Solution. Answer: 14.

We first show that $n \geq 14$. Clearly, it suffices to show $n = 13$ does not work. When $n = 13$, player X can remove card 4.

- If player Y does not remove any of cards 8, 9, 10, then player X can remove cards 8, 9, 10 in the next step.
- If player Y removes cards 7, 8 (or 10, 11 by symmetry), then player X can remove cards 9, 10, 11 in the next step.
- If player Y removes cards 8, 9 (or 9, 10 by symmetry), then player X can remove cards 10, 11, 12 in the next step.

Afterwards, there are no four cards with consecutive labels.

Secondly, we show that $n = 14$ suffices.

- If player X first removes card 1, 2, 5 or 6, then player Y can remove cards 7, 8 in the next step.
- If player X first removes card 3, 4 or 7, then player Y can remove cards 9, 10 in the next step.

Afterwards, there are two disconnected sets of four cards with consecutive labels. The three cards removed by player X in the next step cannot belong to both sets since the sets are disconnected. Therefore, player Y can always complete the final move. $\square$

Problem 2. (JBMO Shortlist 2014 C2) There are n cities in a country, and there are some two-way flights between some pairs of cities. We are planning for a trip to go from one city to another different city by flight. Suppose there are $k > 2014$ possible trips that include no more than one intermediate stop. Find the smallest possible value of n , and find the smallest possible value of k for that value of n . (Note that two trips are different if the routes are different. In particular, the trip going from A to B in one flight is different from the trip going from B to A in one flight.)

Solution. Answer: The smallest n is 14, and the smallest k is 2016.

We construct a simple graph in the obvious way. For each vertex A with degree d , there are d trips of the form $A \rightarrow B$, and $d(d-1)$ trips of the form $B \rightarrow A \rightarrow C$ since there are d choices for B and $d-1$ choices for $C \neq B$. Thus, there are $d + d(d-1) = d^2$ trips of these kinds. Note that all trips are of these kinds where A can be any arbitrary vertex. Therefore, if we let $d_1, d_2, \dots, d_n$ be the degrees of the vertices, then we have

$$k = d_1^2 + d_2^2 + \dots + d_n^2.$$

Using the trivial bound $d_i \leq n-1$, we have $2014 < k \leq n(n-1)^2$. This implies $n \geq 14$. Also, k must be even since we can reverse the trip to create a new trip. This shows $k \geq 2016$.

It remains to provide a possible case where $n = 14$ and $k = 2016$. For example, we label the vertices $A_1, A_2, \dots, A_{14}$ and draw an edge between A_i and A_j whenever $j \neq i-7, i, i+7$. In this simple graph, the degree of each vertex is 12. Therefore, we have $k = 14 \cdot 12^2 = 2016$. $\square$

Problem 3. (Tuymaada Olympiad 2018 Junior Problem 5) There are 99 identical balls, where 50 of them are made of copper, and the other 49 of them are made of zinc. There is a spectrometer test which can be applied to any two balls to determine whether they are made of the same metal or not. However, the result of the test can only be obtained on the next day. If all the tests should be performed today, what is the minimum number of tests required to determine the material of each ball?

Solution. Answer: 98.

We first show that 98 tests is sufficient. For example, we can perform a test between ball 1 and any other ball. There are 98 tests in total, and we can tell whether each ball is of the same kind as ball 1 or not. Thus, we are able to partition the balls into two groups made of the same materials. The group with 50 balls must be made of copper.

Secondly, we show that 97 tests is insufficient. Consider a graph of 99 vertices where each vertex is a ball and two balls are adjacent if we have performed a test between them. WLOG we may assume the graph is simple since it is no use to carry out the same test. If the graph contains a cycle, we can remove any edge in this cycle since we are able to tell the result of this test based on the results of the tests in the same cycle. Then we can draw any other edge which does not create a new cycle (which is possible since the number of vertices is 2 more than the number of edges). Repeating this, we may now assume the graph is a tree. As there are 99 vertices and 97 edges, this tree is composed of two connected components with a and b vertices respectively.

As $a+b=99$, we may assume a is odd while b is even. Let $b=2c$, so that $a=99-2c$. Suppose the result tells us that the first component contains $49-c$ balls of one kind and $50-c$ balls of another kind, and the second component contains c balls of each kind. Then we cannot distinguish between the two kinds of balls in the second component. Therefore, 97 tests is insufficient. $\square$

Problem 4. (European Mathematical Cup 2021 Senior Problem 1) There is a regular polygon with 2021 vertices. We label each vertex by a real number such that the labels of adjacent vertices differ by at most 1. Afterwards, we draw a diagonal between every pair of non-adjacent vertices whose labels differ by at most 1. What is the smallest possible number of diagonals that we draw?

Solution. Answer: 2018

Firstly, we provide an example of drawing exactly 2018 diagonals. Suppose the labels of the vertices are

$$0, 1, 2, \dots, 1009, 1010, 1009.1, \dots, 2.1, 1.1, 0.1$$

in this order. Then only the diagonals joining the pair $(k, k + 0.1)$ and the pair $(k, k - 0.9)$ (where $k = 1, 2, \dots, 1009$) are drawn.

Secondly, we prove that at least 2018 diagonals are drawn. In general, if 2021 is replaced by any $n \geq 4$, we show that at least $n - 3$ diagonals are drawn. When $n = 4$, let the labels be a, b, c, d in this order. WLOG assume a is the smallest label. Then we have $a \leq b \leq a + 1$ and $a \leq c \leq a + 1$. Clearly, $|b - c| \leq 1$, and so there is at least one diagonal joining b and c .

Assume there are at least $n - 3$ diagonals in a polygon of n vertices. Consider a polygon of $n + 1$ vertices, and let the labels be $a_1, a_2, \dots, a_{n+1}$ in this order. WLOG assume a_1 is the smallest label. As above, we know that $|a_2 - a_{n+1}| \leq 1$, and there is a diagonal joining a_2 and a_{n+1} . If we remove the vertex a_1 , the remaining n -gon still satisfies the condition because $|a_2 - a_{n+1}| \leq 1$. By the inductive hypothesis, there are at least $n - 3$ more diagonals in this n -gon. In total we have at least $n - 2$ diagonals in the original polygon of $n + 1$ vertices. This proves the inductive step. So we are done. $\square$

Problem 5. (MEMO 2013 Team Problem 4) There are n points on the plane with no three collinear. Each point is coloured red or blue such that every triangle with vertices of the same colour contains at least one point of the other colour in its interior. Find the largest possible value of n .

Solution. Answer: 8.

Claim. Suppose there are $x \geq 3$ points with no three collinear. Let y be the number of vertices of the convex hull of these points. If we partition this convex hull into triangles using the x points as vertices such that every point is the vertex of some triangles, then there are $2x - y - 2$ triangles in total.

Proof. There are $x - y$ points inside the convex hull. Therefore, the total sum of interior angles of all triangles is

$$(y - 2) \times 180^\circ + (x - y) \times 360^\circ = 180^\circ \times (2x - y - 2).$$

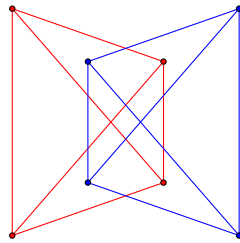
As each triangle contributes 180° , there are $2x - y - 2$ triangles in the partition. $\square$

As a corollary, we can form at least $2x - x - 2 = x - 2$ non-overlapping triangles using x points.

Firstly, suppose $n \geq 9$. By the pigeonhole principle, we may assume there are $k \geq 5$ red points up to symmetry. Let P be the convex hull of the set of all red points, and let m be the number of vertices of P .

By the claim, we can partition P into $2k - m - 2$ red triangles, each of which contains a blue point. Since $2k - m - 2 \geq k - 2 \geq 3$, we can apply the corollary of the claim to show that there are at least $(2k - m - 2) - 2$ non-overlapping blue triangles. Thus, there are at least $2k - m - 4$ red points inside these blue triangles, which must be inside P . As $k \geq 5$, we have $2k - m - 4 \geq k - m + 1$. However, there are only $k - m$ red points inside P , which is a contradiction. So we must have $n \leq 8$.

Secondly, we provide an example with exactly 8 points. This can be easily seen from the figure below.



□

Problem 6. (MEMO 2016 Team Problem 3) In each cell of an 8×8 board, there is at most one house. A house is *in the shade* if there is a house in each of the left, right and bottom adjacent cells. In particular, no house on the leftmost edge, the rightmost edge and the bottommost edge is in the shade. Find the maximum number of houses on the board if no house is in the shade.

Solution. Answer: 50.

Firstly, we show that there are at most 50 houses. It suffices to show that each 8×4 board has at most 25 houses. Consider an 8×4 board. Suppose there are a rows with 4 houses each, b rows with 3 houses each and c rows with at most 2 houses each. For each row with 4 houses, we see that there can be at most 2 houses in the row below (if exists) since the middle 2 cells in that row cannot contain a house. Therefore, we must have $a - 1 \leq c$. It follows that the number of houses is at most

$$4a + 3b + 2c \leq 3a + (c + 1) + 3b + 2c = 3(a + b + c) + 1 = 3 \times 8 + 1 = 25.$$

This proves the upper bound.

Secondly, we provide an example of 50 houses as follows, where X denotes a house.

X	X		X	X		X	X
X	X		X	X		X	X
X	X		X	X		X	X
X	X		X	X		X	X
X	X		X	X		X	X
X	X		X	X		X	X
X	X		X	X		X	X
X	X	X	X	X	X	X	X

□